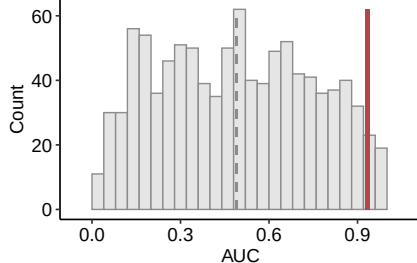


A

Apparent | LM_core

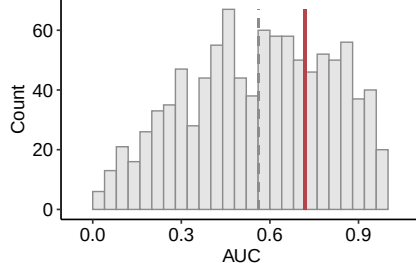
observed 0.933, null 0.489; empirical P = 0.031



B

Apparent | PM_core

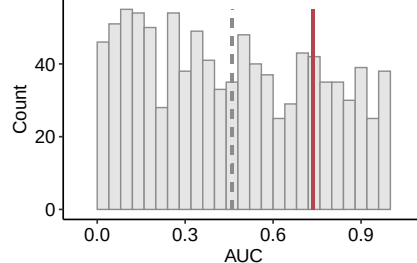
observed 0.719, null 0.561; empirical P = 0.302



C

LOCO main | LM_core (array-derived)

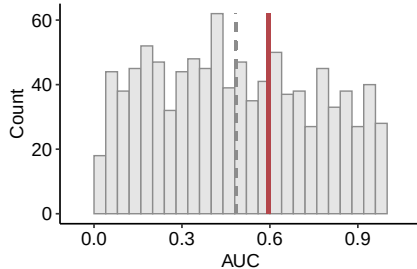
observed 0.736, null 0.460; empirical P = 0.223



D

LOCO restricted | detectable genes

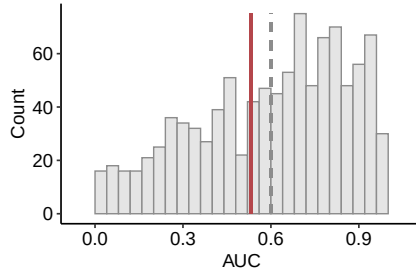
observed 0.596, null 0.484; empirical P = 0.367



E

Cross-cohort | PM_core (GSE190609)

observed 0.532, null 0.600; empirical P = 0.637



Null distributions of the Mann-Whitney AUC for 1,000 random gene sets matched for size and detection rate under each scoring configuration. Solid red line, observed AUC of the true gene set; dashed grey line, random-set mean; empirical P, fraction of random sets with AUC $\geq$ observed. Panel configurations: A: 57 genes, 18 vs 54 PT; B: 32 genes, 54 vs 18 PT; C: 21 genes, 18 vs 54 PT; D: 19 genes, 18 vs 54 PT; E: 73 genes, 19 vs 18 PT.